

1. Introduction

The higher education ecosystem is undergoing a major transformation due to rapid technological advancements, the growth of the digital economy, and continuously changing industry skill requirements. Traditional education models, which are largely centralized and dependent on time-bound degree programs, are increasingly unable to meet the demands of continuous learning and professional upskilling. In modern workforce environments, particularly those influenced by the gig economy, the assumption that a single academic degree is sufficient for a lifetime career is no longer valid

As a result, there has been a growing shift toward competency-based education and the adoption of micro-credentials. Micro-credentials allow learners to acquire and showcase specific skills and competencies, such as technical expertise or domain-specific knowledge, in a modular and flexible manner. These credentials provide a more precise representation of an individual's abilities and align better with current industry needs. However, despite their increasing relevance, existing credentialing infrastructures remain fragmented, inefficient, and heavily centralized, limiting their scalability and trustworthiness.

Traditional credential verification systems involve manual processes, high administrative costs, and long verification timelines. They are also vulnerable to data tampering, certificate fraud, and lack a reliable cross-institutional trust mechanism. These limitations reduce credential portability and weaken trust among students, academic institutions, and employers. Consequently, there is a critical need for a secure, transparent, and decentralized framework that can ensure the authenticity and integrity of academic and professional credentials while supporting seamless verification

In parallel, artificial intelligence has emerged as a powerful tool for skill and job recommendation systems, enabling personalized guidance based on learner profiles and market demands. However, most AI-driven platforms rely on centralized data collection, requiring access to sensitive institutional and student data, which raises serious privacy and compliance concerns. This creates a gap between verified

educational achievements and intelligent, privacy-preserving career and academic recommendations.

To address these challenges, this project presents **Yogyata**, a hybrid decentralized platform that integrates blockchain-based micro-credentialing with AI-powered recommendation systems. The platform utilizes the Polygon blockchain as an immutable ledger for credential anchoring and the InterPlanetary File System (IPFS) for decentralized storage of credential metadata, ensuring scalability and data integrity. Additionally, Yogyata incorporates a Federated Learning framework that enables collaborative intelligence across institutions without sharing raw data, thereby preserving privacy while delivering personalized skill, job, and admission guidance.

By combining decentralized trust, distributed storage, and privacy-preserving artificial intelligence, the proposed system enhances accessibility, trust, and efficiency in digital education systems. This approach directly contributes to the objectives of **Sustainable Development Goal 4 (SDG 4): Quality Education** by empowering learners with secure, portable, and verifiable learning records and supporting informed educational and career pathways

2. Problem Definition

2.1 Problem Statement

The rapid digitization of education and professional skill development has increased the need for reliable, verifiable, and portable academic credentials. However, existing credential management and verification systems are predominantly centralized, fragmented, and inefficient. Educational institutions store credentials in isolated databases, making cross-institutional verification slow, costly, and heavily dependent on manual administrative processes. Verification often takes several business days and requires trusted intermediaries, increasing operational overhead for institutions and employers

These centralized systems are also vulnerable to certificate forgery, unauthorized modifications, and data breaches, which undermines trust in academic and professional credentials. The absence of a tamper-proof and transparent verification mechanism limits credential portability and reduces confidence among stakeholders such as students, employers, and academic bodies. As a result, learners lack full ownership and long-term control over their educational records, restricting the concept of lifelong learning.

In addition to verification challenges, there exists a significant disconnect between verified educational achievements and actionable guidance for career and academic progression. While artificial intelligence is increasingly used for skill and job recommendations, most existing systems require centralized access to sensitive student and institutional data. This raises concerns related to data privacy, regulatory compliance, and ethical use of educational data. Consequently, there is a pressing real-world need for a decentralized, secure, and privacy-preserving framework that can both ensure trustworthy credential verification and provide intelligent, personalized recommendations aligned with evolving workforce demands

2.2 Background Information (Literature Review)

Previous research highlights the potential of blockchain technology to address long-standing issues in academic credentialing. Early blockchain-based micro-credentialing systems demonstrated improvements in transparency, immutability, and automation of certificate issuance and verification. These systems reduced administrative burdens and minimized fraud by leveraging cryptographic primitives, smart contracts, and decentralized ledgers. However, large-scale adoption has been limited due to challenges such as interoperability, institutional resistance, scalability constraints, and lack of standardized frameworks.

To overcome blockchain storage limitations, several studies proposed hybrid architectures that integrate blockchain with decentralized storage solutions such as the InterPlanetary File System (IPFS). In these approaches, credential data or metadata is stored off-chain while cryptographic hashes are anchored on-chain, ensuring integrity without incurring high storage costs. While effective in improving scalability and efficiency, these systems still require robust trust and governance mechanisms for validation and verification.

Recent literature also explores the integration of artificial intelligence with blockchain-based credentialing systems. AI-driven platforms enable personalized learning pathways, skill-gap analysis, and job matching by analyzing learner profiles and market trends. However, most existing AI systems rely on centralized data aggregation, which introduces privacy risks and governance concerns. To address this, Federated Learning has been proposed as a privacy-preserving alternative, allowing collaborative model training across institutions without sharing raw data. Although promising, such integrated blockchain–AI systems remain largely conceptual or limited to small-scale deployments.

Overall, existing studies demonstrate the strong potential of blockchain and AI-enabled micro-credentialing systems but emphasize the need for practical, scalable, and privacy-centric implementations. These gaps in current approaches form the foundation for the proposed Yogyata framework, which aims to combine decentralized trust, distributed storage, and privacy-preserving intelligence into a unified system suitable for modern digital education ecosystems.

3. Objectives

3.1 Primary Objectives

The primary objective of this project is to design and develop a decentralized framework that ensures secure, transparent, and tamper-proof management of academic and professional micro-credentials. The system aims to eliminate reliance on centralized authorities by leveraging blockchain technology as an immutable ledger for credential anchoring and verification. Through this approach, the project seeks to establish a trustworthy mechanism for issuing, storing, and verifying credentials that can be universally accessed and validated.

Another key objective is to provide learners with full ownership and long-term control over their educational records. By enabling credentials to be stored in a decentralized manner and linked to the learner's blockchain identity, the system supports portability and lifelong usability of credentials across institutions and employers. This directly enhances trust and interoperability within the education-to-employment ecosystem.

Additionally, the project aims to integrate intelligent, personalized recommendation capabilities into the credentialing framework. By utilizing a Federated Learning-based artificial intelligence model, the system seeks to deliver skill, course, job, and admission recommendations without accessing or sharing raw institutional or student data. The expected outcome is a privacy-preserving intelligence layer that aligns verified learner competencies with appropriate educational and career pathways.

3.2 Secondary Objectives

A secondary objective of the project is to significantly reduce the time and cost associated with traditional credential verification processes. By automating verification through blockchain smart contracts and decentralized validation mechanisms, the system aims to replace multi-day manual procedures with near real-time verification, thereby lowering administrative overhead for institutions and employers.

The project also aims to enhance data privacy and regulatory compliance within intelligent education systems. Through the use of Federated Learning, the system ensures that sensitive institutional data remains local while still contributing to a globally trained intelligence model. This objective supports ethical AI deployment and compliance with data protection norms.

Another secondary objective is to improve the alignment between educational outcomes and labor market requirements. By performing skill-gap analysis, admission pathfinding, and personalized weighting of learner effort, the system aims to guide learners toward courses, bridge programs, and institutions where they have a higher probability of success. Collectively, these objectives contribute to building a scalable, inclusive, and future-ready digital education ecosystem that supports the goals of quality education and lifelong learning.

4. Methodology

4.1 Approach

The methodology adopted in this project follows a **modular, three-tier decentralized architecture** designed to address issues of trust, scalability, data privacy, and intelligent decision-making in educational credentialing systems. The overall strategy is to decouple data storage, verification, and intelligence into independent yet interconnected layers, ensuring that each concern is handled efficiently without compromising the others.

The first tier, referred to as the **Data Persistence Layer**, focuses on scalable and decentralized storage. Instead of storing credential data directly on the blockchain, which is economically inefficient, the system utilizes the InterPlanetary File System (IPFS). Credential information is serialized, cryptographically hashed, and stored off-chain, while its integrity is preserved through content addressing. This approach ensures immutability and efficient data retrieval without blockchain storage bloat.

The second tier forms the **Trust Protocol Layer**, which acts as the system's source of truth. This layer uses the Polygon blockchain to anchor credential hashes and manage verification states through smart contracts. A customized **Proof-of-Academic-Reputation (PoAR)** mechanism is employed, where credential verification depends on validators weighted by their academic reputation rather than computational power or financial stake. This model aligns consensus with educational governance while preserving the security and efficiency of a Proof-of-Stake blockchain.

The third tier constitutes the **Intelligence Layer**, which integrates artificial intelligence using a **Federated Learning (FL)** framework. The theoretical foundation of this layer is privacy-preserving collaborative learning, where multiple institutions contribute to a shared global model without exposing raw data. Local models are trained on institutional datasets, and only encrypted model updates are aggregated using the Federated Averaging (FedAvg) algorithm. During inference, the trained global model accesses verified, on-chain learner credentials

(with user consent) to generate personalized skill, job, and admission recommendations.

Flow Explanation (Conceptual Flow Chart Description):

1. Credential data is generated after course completion.
2. Data is hashed and stored on IPFS, producing a unique CID.
3. The CID is anchored on the blockchain via a smart contract.
4. Validators verify credentials using PoAR consensus.
5. Verified credentials form a trusted learner skill portfolio.
6. Federated Learning trains a global intelligence model using institutional data.
7. The model generates personalized recommendations based on verified credentials.

This layered approach ensures decentralization, trust, scalability, and intelligent personalization within a unified framework

4.2 Procedures

The execution of the project followed a structured sequence of procedures, ensuring systematic development and validation of each component.

Initially, credential data was defined in a standardized format, including learner identifiers, course information, and completion timestamps. This data was serialized into JSON format and cryptographically hashed using SHA-256 to ensure integrity. The hashed credential payload was then uploaded and pinned to IPFS using the Pinata gateway, generating a unique Content Identifier (CID).

In the next step, smart contracts were developed and deployed on the Polygon Amoy test network. These contracts stored minimal credential metadata, including the CID, credential owner address, and verification status. Validators interacted

with the contract to report and verify credentials, with verification status updated only after achieving consensus through the Proof-of-Academic-Reputation mechanism.

Parallel to credential verification, the Federated Learning framework was established. Educational institutions trained local models on private historical datasets, such as student performance and admission trends. Only encrypted weight updates were shared with a central aggregator, which applied the Federated Averaging algorithm to update the global model. This process was repeated iteratively to improve recommendation accuracy while maintaining data privacy.

During the inference phase, learners granted permission for the system to access their verified blockchain credentials. The global model analyzed the learner's verified skill vector to generate personalized recommendations, including skill-gap analysis, bridge course suggestions, and admission pathfinding.

Milestones included successful IPFS integration, smart contract deployment, validator consensus testing, federated model convergence, and end-to-end system evaluation. Each stage was validated before proceeding to the next, ensuring reliability and correctness of the overall system

5. Project Execution

5.1 Planning and Design

The planning and design phase focused on identifying the core limitations of existing centralized credentialing and recommendation systems and translating these challenges into a decentralized, privacy-preserving solution. Initial brainstorming centered on how to ensure credential authenticity, reduce verification delays, and maintain learner ownership while supporting intelligent recommendations without compromising institutional data privacy.

Based on this analysis, a **modular three-tier architecture** was conceptualized during the design stage. The design deliberately separated data storage, trust management, and intelligence into independent layers to avoid system bottlenecks and ensure scalability. Early design drafts emphasized minimizing on-chain data storage to reduce costs, which led to the decision to use IPFS for credential metadata and the blockchain only for anchoring cryptographic proofs.

Another key design decision involved the selection of an appropriate trust mechanism. Instead of traditional Proof-of-Work or purely financial stake-based consensus, the system design incorporated a **Proof-of-Academic-Reputation (PoAR)** model. This choice was driven by the need to align credential verification with academic governance, where trust is derived from institutional reputation rather than computational or monetary power.

For the intelligence component, the design phase identified privacy as a critical requirement. To address this, a Federated Learning framework was selected, enabling collaborative model training across institutions without sharing raw data. The design also included a clear separation between training and inference phases, ensuring that sensitive institutional data remained private while verified learner credentials could be used for personalized recommendations. These design considerations collectively shaped the final system architecture prior to implementation.

5.2 Implementation

The implementation phase involved translating the finalized system design into a working decentralized software prototype. Development began with the **Data Persistence Layer**, where credential data was structured, serialized, and cryptographically hashed before being uploaded to IPFS using the Pinata SDK. The resulting Content Identifiers (CIDs) were retained for blockchain anchoring, ensuring content integrity and tamper detection.

Subsequently, the **Trust Protocol Layer** was implemented through smart contracts written in Solidity and deployed on the Polygon Amoy test network. The smart contracts managed credential registration, verification states, and validator interactions. Special attention was given to gas optimization by storing only essential metadata on-chain, such as CIDs and verification status, while keeping bulk data off-chain. Validators interacted with the contracts to report and verify credentials, with verification finalized only after consensus through the PoAR mechanism.

In parallel, the **Intelligence Layer** was implemented as a simulated Federated Learning environment. Local training procedures were executed at institutional nodes using private datasets, and only encrypted model updates were shared with a central aggregator. The Federated Averaging algorithm was applied to generate a global model capable of understanding skill-to-job mappings and admission trends. During inference, the system accessed verified blockchain credentials (with user permission) to generate personalized recommendations using the trained global model.

The frontend prototype was developed using a React-based architecture integrated with blockchain interaction libraries to allow users to authenticate, manage credentials, and view recommendations. Each component was tested independently before being integrated into an end-to-end workflow. This phased execution ensured that the planned design objectives were systematically realized in the final working prototype.

6. Tools and Techniques Used

6.1 Tools

The project utilized a combination of blockchain platforms, decentralized storage systems, software frameworks, and development tools to implement the proposed decentralized credentialing and recommendation system.

Polygon Amoy Test Network

The Polygon Amoy blockchain was used as the underlying decentralized ledger for anchoring credential records. It was chosen due to its low transaction latency, minimal gas costs, and full compatibility with the Ethereum Virtual Machine (EVM). The blockchain served as the immutable layer of trust, storing credential references and verification states through smart contracts.

Solidity (v0.8.20)

Solidity was used to develop smart contracts that manage credential registration, verification status, and validator interactions. The contracts ensured that only authorized validators with sufficient academic reputation could participate in credential verification, implementing the Proof-of-Academic-Reputation mechanism.

InterPlanetary File System (IPFS)

IPFS was employed for decentralized storage of credential metadata. Instead of storing large credential files on-chain, IPFS enabled content-addressed storage where any modification to data results in a different hash, ensuring data integrity and scalability.

Pinata SDK

The Pinata platform was used as an IPFS gateway to pin and manage credential files reliably. It facilitated efficient uploading, retrieval, and persistence of credential data while maintaining decentralization.

React.js with Vite

The frontend of the system was developed using React.js with the Vite build tool to create a responsive and modular user interface. This allowed users to interact

with the blockchain, manage credentials, and access personalized recommendations.

Ethers.js

Ethers.js was used to enable communication between the frontend application and the blockchain. It handled wallet connections, smart contract interactions, and transaction execution in a secure and user-friendly manner.

Supabase

Supabase was used for authentication and role-based access management. It helped maintain session persistence and distinguish between different user roles such as students and administrators.

Federated Learning Simulation Environment

A simulated Federated Learning environment was used to model distributed training across multiple institutional nodes. This environment supported local model training, encrypted gradient sharing, and centralized aggregation without exposing raw data.

6.2 Techniques

Several techniques and methodological approaches were applied to ensure security, scalability, and privacy in the proposed system.

Decentralized Credential Anchoring

Instead of centralized certificate storage, the system anchored credential references on the blockchain. This technique ensured immutability, transparency, and tamper resistance, enabling instant and trustworthy verification.

Content Addressing and Cryptographic Hashing

Credential data was serialized and hashed using SHA-256 before being stored on IPFS. This technique ensured data integrity, as any modification to the credential data would result in a different content identifier.

Proof-of-Academic-Reputation (PoAR)

The project employed a reputation-based validation technique where credential verification depended on validators weighted by academic reputation rather than computational power or financial stake. This approach was chosen to align consensus mechanisms with educational governance and reduce fraud risks.

Federated Learning with FedAvg Algorithm

Federated Learning was used to train a global recommendation model across institutions without sharing raw data. Local models were trained on private datasets, and only encrypted model updates were aggregated using the Federated Averaging algorithm. This technique ensured privacy preservation while enabling collaborative intelligence.

Skill Gap Analysis and Similarity Measurement

Mathematical techniques such as Euclidean distance and cosine similarity were applied to compare a learner's verified skill profile with job or institutional requirements. These methods enabled accurate identification of skill gaps and appropriate bridge courses.

Personalized Weighting Algorithm

A personalized weighting approach was applied to rank recommendations based on verified effort, skill gaps, and learner preferences. This technique improved recommendation relevance by prioritizing effort-based and interest-aligned outcomes over generic popularity-based suggestions.

These tools and techniques collectively enabled the development of a secure, scalable, and privacy-preserving platform for decentralized micro-credentialing and intelligent educational guidance

7. Partial Results

7.1 Initial Findings

During the early stages of implementation and testing, several important findings were observed that validated the feasibility of the proposed decentralized architecture. Initial deployment of the credential anchoring mechanism on the Polygon Amoy blockchain demonstrated that credential verification could be completed within seconds, as opposed to the multi-day timelines commonly associated with traditional, centralized verification systems. This confirmed the effectiveness of using blockchain as a real-time trust layer for academic and professional credentials.

Early experiments with decentralized storage revealed that storing credential metadata on IPFS significantly reduced storage costs while maintaining data integrity. By anchoring only cryptographic hashes on-chain and storing actual credential data off-chain, the system avoided blockchain storage bloat and achieved efficient retrieval without compromising security. Initial integrity checks confirmed that any modification to credential data resulted in a different content identifier, ensuring tamper detection.

Preliminary testing of the Federated Learning framework showed that collaborative model training across institutional nodes was possible without sharing raw datasets. The initial global model was able to capture basic patterns related to skill-to-job mappings and admission trends while preserving institutional data privacy. These early findings indicated that privacy-preserving intelligence could be successfully integrated with decentralized credential systems.

7.2 Iterative Improvements

Based on the initial findings, several iterative refinements were introduced to improve system reliability, accuracy, and user relevance. One of the key improvements involved strengthening the validator verification mechanism. The Proof-of-Academic-Reputation (PoAR) model was refined to more accurately weight validator contributions by incorporating factors such as historical verification success, consensus agreement, and audit-based accuracy. This iteration improved resistance to fraudulent or incorrect credential verification.

Improvements were also made to the recommendation ranking logic within the intelligence layer. Initial recommendation outputs tended to prioritize generic skill matches. To address this, a personalized weighting approach was introduced, incorporating verified learner effort (such as recorded hours spent) alongside skill gaps and user interests. This adjustment significantly improved the relevance of recommendations by favoring effort-based and goal-aligned outcomes.

Additionally, iterative tuning of the Federated Learning aggregation process enhanced model convergence and recommendation accuracy. By adjusting aggregation weights and refining local update calculations, the global model became more effective at identifying appropriate bridge courses and admission pathways. These iterative enhancements were guided directly by early-stage observations and performance evaluations, ensuring continuous improvement of the overall system.

8. Results and Discussion

8.1 Final Results

The final evaluation of the proposed Yogyata system was conducted across multiple dimensions, including verification performance, storage efficiency, cost effectiveness, recommendation accuracy, and data privacy. Experimental benchmarking demonstrated a substantial improvement over traditional credential management systems.

Credential verification latency was reduced from several business days in conventional systems to approximately **15–30 seconds** using the blockchain-based verification process. This improvement was achieved through automated smart contract execution and decentralized validation on the Polygon network. The results confirmed that blockchain anchoring enables near real-time credential verification without reliance on intermediaries

In terms of storage efficiency, the use of IPFS for credential metadata significantly reduced costs. While centralized systems incur notable storage expenses for hosting credential files, the decentralized IPFS-based approach achieved storage costs of **less than \$0.01 per MB**, representing a substantial reduction. On-chain anchoring costs were also minimal, with gas fees averaging approximately **\$0.002 per transaction**, demonstrating the economic viability of the system.

The Federated Learning–based recommendation engine was evaluated through simulation. The global model successfully learned skill-to-job mappings and institutional admission patterns without accessing raw institutional data. The **admission pathfinding module achieved an accuracy of approximately 94%**, effectively identifying suitable bridge courses and educational pathways based on verified learner profiles. Additionally, personalized weighting based on verified effort improved recommendation relevance, with a majority of users favoring effort-aware recommendations over generic alternatives.

From a security perspective, the system demonstrated strong resilience against common threats. Data integrity was ensured through cryptographic hashing, and no data leakage occurred during AI training since only encrypted model updates were

shared. Validator participation was regulated through reputation staking, reducing susceptibility to Sybil and malicious verification attempts

8.2 Discussion

The results indicate that the primary objectives of the project were successfully achieved. The decentralized credentialing framework effectively addressed the challenges of trust, verification delay, and credential portability by leveraging blockchain immutability and decentralized storage. The drastic reduction in verification time and operational cost highlights the system's practical relevance for real-world educational and employment contexts.

The integration of Federated Learning proved particularly significant, as it enabled intelligent, personalized recommendations while maintaining strict data privacy. This validates the feasibility of combining decentralized trust mechanisms with privacy-preserving artificial intelligence in education systems. The high accuracy achieved in admission pathfinding demonstrates the potential of verified micro-credentials to meaningfully guide learners toward suitable academic and career opportunities.

An important observation was the strong impact of incorporating verified learner effort into recommendation ranking. The preference for effort-based recommendations suggests that learners value guidance that reflects their demonstrated commitment rather than purely popularity-driven metrics. This outcome reinforces the importance of transparent, verifiable data in intelligent decision-making systems.

Overall, the findings confirm that the proposed Yogyata framework not only meets its intended objectives but also establishes a scalable and privacy-centric model for future digital education ecosystems. The results demonstrate that decentralized micro-credentialing combined with AI-driven intelligence can significantly enhance accessibility, trust, and quality in education, aligning closely with the goals of sustainable and lifelong learning

9. Prototype (Software)

9.1 Prototype Description

The developed prototype is a **software-based decentralized platform** that enables secure issuance, storage, verification, and intelligent utilization of academic and professional micro-credentials. The system integrates blockchain technology, decentralized storage, and privacy-preserving artificial intelligence into a unified working model.

The prototype consists of three tightly integrated layers. The **Data Persistence Layer** uses the InterPlanetary File System (IPFS) to store credential metadata in a decentralized manner. Credential data is serialized and cryptographically hashed, producing a unique Content Identifier (CID) that guarantees integrity and tamper detection. This design ensures scalability while avoiding the high costs associated with on-chain storage.

The **Trust Protocol Layer** is implemented on the Polygon Amoy blockchain using Solidity-based smart contracts. These contracts maintain credential ownership, verification status, and validator interactions. A Proof-of-Academic-Reputation (PoAR) mechanism governs credential verification, where validators stake academic reputation rather than computational power or financial assets. This layer provides an immutable and transparent source of truth for credential verification.

The **Intelligence Layer** incorporates a Federated Learning–based recommendation system. The prototype supports skill, job, and admission recommendations by analyzing verified learner credentials stored on the blockchain. A personalized weighting algorithm ranks recommendations based on skill gaps, verified effort, and learner preferences. The frontend, built using a React-based architecture, enables users to authenticate, manage credentials, and view personalized recommendations through a responsive interface. Collectively, these features demonstrate a functional decentralized credentialing and recommendation platform

9.2 Development Process

The prototype development followed a phased and modular approach to ensure reliability and alignment with the system design. Development began with defining the credential data structure and implementing cryptographic hashing and IPFS integration using the Pinata SDK. This phase focused on ensuring content addressing, integrity verification, and reliable off-chain storage.

Subsequently, smart contracts were developed in Solidity and deployed on the Polygon Amoy test network. The development process emphasized minimizing on-chain data to reduce gas costs while maintaining essential verification functionality. Implementing the Proof-of-Academic-Reputation mechanism required careful design of reputation tracking, validator eligibility checks, and consensus enforcement.

In parallel, the Federated Learning framework was implemented in a simulated environment. Local training logic was executed at institutional nodes, and encrypted model updates were aggregated using the Federated Averaging algorithm. One of the key challenges was ensuring effective model convergence without access to centralized datasets. This was addressed by iteratively refining aggregation weights and local update calculations.

Frontend development was carried out using React and integrated with blockchain interaction libraries to enable seamless wallet connections and contract interactions. Integration challenges mainly involved synchronizing on-chain credential states with off-chain recommendation outputs. These challenges were resolved through modular component design and staged integration testing across layers

9.3 Testing and Validation

Testing and validation of the prototype were conducted across functionality, performance, security, and recommendation accuracy dimensions. Functional testing verified correct credential issuance, IPFS storage, blockchain anchoring, and validator-based verification workflows. Each smart contract function was tested to ensure proper access control, state transitions, and event handling.

Performance testing focused on verification latency, storage cost, and transaction fees. Results showed that credential verification could be completed within seconds and that storage and gas costs remained minimal due to the hybrid on-chain/off-chain design. These results validated the efficiency and scalability of the prototype.

The Federated Learning–based recommendation system was validated through simulation. Testing confirmed that the global model could generate meaningful skill, job, and admission recommendations without accessing raw institutional data. Admission pathfinding achieved high accuracy, and personalized weighting improved recommendation relevance. Security validation demonstrated resistance to data tampering through cryptographic hashing and reduced susceptibility to fraudulent verification through reputation-based validator staking.

Overall, testing and validation results confirmed that the prototype met its intended functional and performance objectives and effectively demonstrated the feasibility of a decentralized, privacy-preserving credentialing and recommendation system

10. Conclusion

10.1 Summary

This project addressed the growing challenges associated with centralized educational credentialing systems, including lack of trust, high verification latency, limited portability, and vulnerability to fraud. Traditional mechanisms for issuing and verifying academic and professional credentials were found to be inefficient and poorly aligned with the needs of modern, skill-driven education and employment ecosystems. Additionally, existing AI-based recommendation systems often compromise data privacy by relying on centralized access to sensitive institutional and learner data.

To overcome these limitations, the project proposed and implemented **Yogyata**, a decentralized platform that integrates blockchain-based micro-credentialing with AI-powered, privacy-preserving recommendation systems. The system leveraged the Polygon blockchain to serve as an immutable trust layer for credential anchoring and verification, while the InterPlanetary File System (IPFS) was used for decentralized and scalable storage of credential metadata. A Proof-of-Academic-Reputation (PoAR) mechanism ensured that credential verification was aligned with academic governance rather than purely financial or computational factors.

The project successfully met its primary objectives by enabling secure, transparent, and near real-time credential verification, significantly reducing verification delays and operational costs. Learners were provided with greater ownership and portability of their credentials through decentralized storage and blockchain anchoring. The integration of a Federated Learning–based intelligence layer further achieved the objective of delivering personalized skill, job, and admission recommendations without exposing raw institutional or student data.

Final results demonstrated substantial improvements over traditional systems, including drastic reductions in verification time, lower storage and transaction costs, high accuracy in admission pathfinding, and complete prevention of data leakage during AI training. These outcomes validate the effectiveness of

combining decentralized trust mechanisms with privacy-preserving artificial intelligence.

Overall, the project demonstrates that blockchain-based micro-credentials, when combined with federated AI models, can form a scalable, secure, and intelligent digital education ecosystem. The proposed framework contributes meaningfully toward improving accessibility, trust, and quality in education, thereby aligning closely with the objectives of **Sustainable Development Goal 4: Quality Education**

10.2 Personal Reflection

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